

Yingru (Richard) Li

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Reinforcement Learning · LLM · Optimization · Reasoning & Agents

EXPERIENCE

xAI - SpaceX

Member of Technical Staff

Singapore

December 2025 – Present

- Drive large-scale (“big-run”) reinforcement learning for the Grok models (Grok 4.x, Grok Build), spanning both the **science** and the **infrastructure** of RL training.
- **Science of RL** — stability, efficiency, and the core algorithms and recipes; identified and resolved critical training instabilities that benefit every stage of the RL pipeline, directly enabling the longest stable RL run.
- **Long-horizon RL** — developing recipes for higher intelligence per token and recursive self-improvement: RL training with state compaction, self-distillation for dense credit assignment, beyond 1 bit per rollout.
- **RL infrastructure** — built and scaled the systems behind these recipes, including big-run training across $\gg 10,000$ GPUs.

ByteDance

LLM Algorithm Research Scientist

Singapore

2025 – December 2025

- Led research on RL training stability and efficiency for LLM reasoning and agents.
- Diagnosed and formalized the **RL Collapse** phenomenon, tracing instability to implicit off-policy-ness caused by the training–inference mismatch (which induces gradient variance and bias); developed **Trust Region Masking**, **Dynamic Vocabulary Pruning**, the **Optimal Token Baseline**, and a simple LR-scheduling fix — complementary methods that together address these root causes.
- Co-led **SimpleTIR** (end-to-end multi-turn tool-integrated reasoning) and **Dr. Kernel** (Triton kernel generation agent via RL).

Tencent AI & Robotics X

Student Research Staff Member, Agent Center

Shenzhen, China

2019 – 2022

- Led **HyperAgent** (ICML 2024) and **HyperDQN** (ICLR 2022) for scalable exploration in deep RL, and **Divergence-Augmented Policy Optimization** (NeurIPS 2019) for stable off-policy data reuse.

Microsoft Research Lab – Asia

Research Intern, Theory Center

Beijing, China

2016

- Research on combinatorial optimization and reinforcement / bandit learning algorithms.

SELECTED PUBLICATIONS

Full list at [Google Scholar](https://scholar.google.com/citations?user=szrlee). ★ denotes equal contribution; † denotes corresponding author.

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**

Yingru Li, Jiawei Xu, Ziniu Li, Jiakai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026*.

2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**

Yingru Li, Jiakai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026*.

3. **Optimistic Thompson Sampling for No-Regret Learning in Unknown Games**

Yingru Li, Liangqi Liu, Wenqiang Pu, Hao Liang, Zhi-Quan Luo. *IEEE Transactions on Signal Processing*, 2026.

4. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**

Wei Liu, Jiawei Xu, Yingru Li, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026*.

5. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**

Jiawei Wang, Jiakai Liu, Yuqian Fu, Yingru Li, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *ICML 2026*.

6. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**

Zhenghai Xue, Longtao Zheng, Qian Liu, Yingru Li, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026*.

7. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
Jiacai Liu*, **Yingru Li***[†], Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Blog*, 2025.
8. **Scalable Exploration via Ensemble++**
Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS* 2025.
9. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**
Yingru Li, Jiawei Xu, Jiakai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv* 2025.
10. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
Meng kang Hu, Yuhang Zhou, Wendong Fan, ..., **Yingru Li**, ..., Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS* 2025.
11. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML* 2024.
12. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
Yingru Li, Zhi-Quan Luo. *AISTATS* 2024.
13. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
Ziniu Li, **Yingru Li**[†], Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR* 2022.
14. **Divergence-Augmented Policy Optimization**
Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS* 2019.

EDUCATION

The Chinese University of Hong Kong (CUHK) 2025

Ph.D. in Computer Science and Information Engineering.

Advisor: *Zhi-Quan (Tom) Luo*. Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST) 2020

B.E./M.S. in Computer Science. Ranked 1st/134 overall; 1st/26 in Computer Theory and Software.

SELECTED HONORS

- National Scholarship (China) — highest national academic honor, top 0.2% nationwide, 2018.
- Best Student Paper, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- SRIBD Ph.D. Fellowship (Gold Class), 2023; Presidential Ph.D. Fellowship, CUHK, 2019–2023.
- Qiming Star Award — 1 of 5 recipients from 7,112 undergraduates, HUST, 2016.